

Chapter

String Handling

1. Write the output of the below code: `String t = "Reduce Reuse Recycle";
String x = new String(t); System.out.print(x == t);` MAIN 2024

- (a) True
- (b) False
- (c) Error
- (d) 1

2. Write the output of the below code: `String t = "Reduce Reuse Recycle";
String x = new String(t); System.out.print(x.equals(t));` SQP 2023

- (a) True
- (b) False
- (c) Error
- (d) 1

3. What is the result of `indexOf("a")` for the string "banana"? COMP 2021

- (a) 0
- (b) 1
- (c) 2
- (d) 3

4. Which method replaces all occurrences of a character in a string?
SQP 2023

- (a) `replace()`
- (b) `replaceAll()`
- (c) `substring()`
- (d) `charAt()`

5. Consider the following program segment to find the length of a string. The statements are jumbled. Choose the correct order of statements. SQP 2022

```
void findLength (String str) {  
    int length;           // (1)  
    System.out.println(length); // (2)  
    length = str.length (); // (3)  
}
```

- (a) (1) (3) (2)
- (b) (3) (1) (2)
- (c) (2) (1) (3)
- (d) (1) (2) (3)

6. Write a statement each to perform the following tasks on a string: SQP 2024

- (i) Find and display the position of the last space in a string s
- (ii) Convert a number stored in a string variable x to double data type.

7. Give the output of the following: COMP 2024

```
String n = "Computer Knowledge";  
String m = "Computer Applications";  
System.out.println(n.substring(0,8).concat(m.substring(9)));  
System.out.println(n.endsWith("e"));
```

8. Write the output of the following: MAIN 2021

- (i) System.out.println(Character.toUpperCase(R));
- (ii) System.out.println(Character.toUpperCase(J));

9. State the purpose and return data type of the following String functions: (i) indexOf()

(ii) compareTo() **COMP 2023**

10. State the output when the following program segment is executed:

```
String a = "Smartphone", b = "Graphic Art";  
String h = a.substring (2, 5);  
String k = b.substring (8).toUpperCase ();
```

System.out.println (h);

System.out.println(k. equalsIgnoreCase(h));

COMP 2022

11. Explain briefly how Java manages strings?

COMP 2024

12. Write the two ways in which you can assign a text string "TOGETHER" to a string variable, say, str .

MAIN 2023

13. Define a method replace Substring(String input) to replace the first occurrence of the word "apple" with "orange". The modified string should be returned to the main method, where it is displayed.

SQP 2024

14. Write a Java method swapCase(String input1, String input2) that converts the first string (input1) to uppercase and the second string (input2) to lowercase. Display the converted results.

SQP 2023

15. Write a Java program to take a string in lowercase and convert the first letter of each word to uppercase. Display the modified string. Sample Input: we are in cyber world Sample Output: We Are In Cyber World.

COMP 2022

Solution

1. (b) False

The == operator compares references, not content. Here, t refers to a string literal, while x is a new String object. Since their references are different, x == t is false.

2. (a) True

The equals() method compares the content of strings, not their references. Since both x and t contain "Reduce Reuse Recycle", x.equals(t) returns true.

3. **Correct answer: (b) 1**

In the string "banana", indexing starts from 0.

The first

4. **Correct answer: (a) replace()**

replace() replaces **all occurrences** of a character or substring in a string.

5. (a) (1) (3) (2)

Correct order:

1. Declare the variable
2. Assign the string length
3. Print the length

6. Here are the required statements:

(i) **Find and display the position of the last space**

```
System.out.println(s.lastIndexOf(" "));
```

(ii) **Convert a number stored in string variable x to double**

```
double d = Double.parseDouble(x);
```

7. ComputerApplications true

8. (i) true

(ii) J

9. (i) indexOf()

- **Purpose:** Finds the index of the first occurrence of a character or substring in a string.
- **Return type:** int

(ii) compareTo()

- **Purpose:** Compares two strings lexicographically.
- **Return type:** int

10. Here, h = art and k = ART Output: art true

11. Java manages strings using **String** and **StringBuffer** classes. A String in Java is an **object**, not a character array. Strings are **immutable**, meaning their value cannot be changed once created. For operations that require modification, Java uses **StringBuffer** (or **StringBuilder**), which is mutable.

Java provides many built-in methods for string manipulation, making handling text easier. All data types can be converted to strings, and numeric strings can be converted back to numbers.

12. Here are the **two correct ways** to assign the text string **"TOGETHER"** to a string variable **str**:

(i) `String str = "TOGETHER";`

(ii) `char arr[] = {'T','O','G','E','T','H','E','R'};`

`String str = new String(arr);`

13.

```
public class Replace Substring {
    // Define the method to replace
    the first occurrence of "apple" with
    "orange"
        public static String
replaceSubstring(String input) {
    // Replace the first occurrence
    of "apple" with "orange"
        return input.replaceFirst("\\
bapple\\b", "orange");
    }
public static void main(String[]
args) {
    // Example input string
    String input = "An apple a
day keeps the doctor away. The apple
is sweet.";
    // Call the replaceSubstring method
    String modifiedString =
replaceSubstring(input);
    // Display the original and modified
strings
        System.out.println("Original
String: " + input);
        System.out.println("Modified
String: " + modifiedString);
    }
}
```

14.

```
public class SwapCaseExample {
    public static void swapCase(String
input1, String input2) {
        // Convert the first string to
uppercase
        String upperCaseStr = input1.
toUpperCase();
        // Convert the second string
to lowercase
        String lowerCaseStr = input2.
toLowerCase();
        // Display the results
        System.out.println("First string
in uppercase: " + upperCaseStr);
        System.out.println("Second string
in lowercase: " + lowerCaseStr);
    }
    public static void main(String[]
args) {
        // Example usage
        swapCase("hello World", "JAVA
Programming");
    }
}
```

15.

```
import java.util.*;
class Capitalise {
    public static void main() {
        boolean flag = false;
        System.out.println("enter a
sentence in lower case to capitalise
first letter of each word");
        Scanner sc = new Scanner(System.
in);
        String g = sc.nextLine();
String s = g.trim();
String n = "";
char ch = Character.toUpperCase(s.
charAt(0));
n = n + ch;
for (int i = 1; i <= s.length() - 1;
i++) {
    if ((s.charAt(i) == ' ') &&
(s.charAt(i + 1) != ' ')) {
        ch = Character.toUpperCase(s.
charAt(i + 1));
        n = n + ch;
    } else {
        ch = s.charAt(i);
        n = n + ch;
    }
}
n = n + s.charAt(s.length() - 1);
System.out.println(n);
}
}
```